SiC Games: Project Specification - Carbon Prototype (V1.3)

I. Background & Framework: "The Active Way"

The **SiC Games** framework is a generative social simulation designed to test the **Stasis vs. Expansion** hypothesis. Traditional modeling often focuses on rational agents that optimize for efficiency, leading to a "Perfect Manual" state—a frozen, low-entropy equilibrium.

This project emulates **Carbon-based (C) population dynamics**, characterized by high-entropy, low-fidelity, and status-driven behaviors. The central framework posits that "flaws"—such as ego, social stratification, and decision noise—are functional mechanisms that drive a population to tunnel through activation barriers and occupy diverse ecological niches.

The Core Emulation

- **The Environment:** A rugged fitness landscape of varying resource density and activation energy (E_a).
- **The Agents:** High-energy, short-lived units motivated by metabolic survival and social prestige.
- **The Goal:** Observe the emergence of **Social Turbulence** as a survival strategy.

II. C-Pop Dynamics: Traits, Factors, & Rationale

Trait/Factor	Description	Rationale & Literature Basis
Ego-Noise (σ)	Decision uncertainty scaling with social status.	Prevents local optima traps. *Roll (1986)* identifies hubris as a driver for high-variance risk-taking.
Social Cred ($\mathcal{C}$)	Non-metabolic currency gained via task success.	"Social Potential Energy" for aggregation. *Dunbar (1992)* suggests social structures have cognitive thresholds.
The Matthew Effect	Proportional reward distribution.	Drives stratification. *Merton (1968)* observed prestige facilitates further resource acquisition.
Legacy Transfer	Continuous resource pumping from parent to progeny.	Lowering "Infant Mortality." *Kaplan et al. (2000)* show human survival depends on extended subsidies.
Heuristic Drift	Lossy transfer of logic manuals.	Prevents static perfection. *Boyd & Richerson (1985)* argue imperfect transmission maintains flexibility.

III. Mathematical Framework

1. Agent State Vector (S_i)

$$S_i = \{ \vec{x}, \mathcal{R}, \mathcal{C}, \phi, H, \text{age} \}$$

- $\vec{x}$: Spatial coordinates.
- $\mathcal{R}, \mathcal{C}$: Resource and Cred buffers.
- ϕ: Genetic drive (Rationalist $[0] \leftrightarrow$ Pretender $[1]$).
- H: Heuristic logic vector (task-solving weights).

2. Decision Logic (Softmax-Langevin)

The probability of agent i choosing task j: $$P(\text{task}_j) = \frac{e^{\frac{U_{i,j}}{\sigma(\mathcal{C}_i)}}}{\sum e^{\frac{U_{i,k}}{\sigma(\mathcal{C}_i)}}}$$

- **Utility (U):** $U = w_R \cdot \nabla \mathcal{R} + w_C \cdot \nabla \mathcal{C}$.
- **Ego-Noise (σ):** $\sigma(\mathcal{C}_i) = \sigma_{\text{base}} + \kappa \cdot \mathcal{C}_i$.

3. Task Resolution (Matthew Rule)

Effective Activation Barrier for a cluster of N agents: $$E_a^{\text{eff}} = E_a \cdot e^{-\gamma \cdot \sum \mathcal{C}_i}$$ **Reward Partition:** $$\Delta \mathcal{R}_i = \mathcal{R}_{\text{tot}} \times \frac{\mathcal{C}_i}{\sum \mathcal{C}_j} \quad , \quad \Delta \mathcal{C}_i = \mathcal{C}_{\text{tot}} \times \frac{\mathcal{C}_i}{\sum \mathcal{C}_j}$$

IV. Parameter Calibration & Stability Window

Parameter	Sim Value	Real-World Basis / Estimation
Lifespan (τ)	400 - 600 steps	~40-50 years at 1 step/month (*Leonard & Robertson, 1997*).
Regrowth (ρ)	0.5% - 2.0%	Net Primary Productivity recovery rates (*Field et al., 1998*).
Cred Decay (δ)	0.5% - 2.0%	Decay of collective/personal memory (*Candia et al., 2019*).
Legacy Transfer	10% - 30%	Parental net energy subsidy to offspring (*Kaplan et al., 2000*).
Ego Coupling (κ)	0.1 - 0.5	20–40% increase in variance in successful leaders (*Roll, 1986*).
Matthew Power	~ 1.5	Standard Pareto wealth/prestige distribution (*Gabaix, 2009*).

V. Environmental Topography: The World Map

The 100 worlds are distributed along a **Connectivity Axis** using Fractal (Perlin) Noise:

- **Resource** R_{low}: A seasonal traveling wave (Sine oscillation) forcing nomadic movement.
- **Resource** R_{high}: Stationary "Mines" (High E_a) and migratory "Great Beasts" (Random Walk).
- **Topography:** Ranges from **Pangea** (high connectivity) to **Archipelago** (isolated pockets).

VI. Implementation Hierarchy

Level 1: Stability Window & Tuning (Current Priority)

Goal: Identify the "Goldilocks Zone" where the system avoids both totalitarian "God-Emperors" and communal stagnation.

- **Focus:** Find the κ, δ, γ values that allow high E_a task resolution without triggering total desertification.
- **Benchmark:** Stable Gini Coefficient between **0.3 and 0.6**.

Level 2: Entropy & Biology (The "Humanity" Prototype)

Goal: Introduce aging, biparental recombination, and heuristic drift.

- **Benchmark:** Emergence of "Sawtooth" population volatility and allopatric logic speciation.

Level 3: Continental Scale (The 100-World Run)

Goal: Full deployment across the Pangea-Archipelago spectrum.

- **Benchmark:** Proof that Carbon logic occupies more niche space and tunnels through higher "Innovation Bandgaps" than Silicon controls.

VII. Coding Strategy for Robustness

- **Data Strategy:** Use **Global State Matrices** (Flat arrays) rather than Object-Oriented agents to maximize CPU cache performance and vectorization.
- **Spatial Optimization:** Implement **Spatial Hashing** (Grid lookup) to reduce neighbor search from $O(N^2)$ to $O(N)$.
- **Clamping:** Hard clamps on σ to prevent numerical instability/explosive ego jumps.

VIII. Success Benchmarks: The "Humanity" Signature

1. **Turbulence:** Non-monotonic population fluctuations.
2. **Stratification:** Clear, functional hierarchies (Gini 0.3–0.6).
3. **Speciation:** Distinct logical branches (H) in isolated regions.
4. **Expansion:** Occupation of sub-optimal territories driven by status-noise.